

Mini-Activity-1-Solutions.R

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```
##### Mini-Assignment 1 Solutions #####
```

```
# Load libraries
```

```
library(psych)
```

```
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.1.5
```

```
## v forcats    1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.1      v tibble    3.3.1
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x ggplot2::%+%( ) masks psych::%+%( )
```

```
## x ggplot2::alpha() masks psych::alpha()
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

1) Import the TIPI_data.csv file into R and save it as a dataframe called personality NOTE: I use read_csv() from tidyverse, rather than read.csv(), since read_csv() automatically converts the imported data to a tibble, which I prefer.

```
# Import data
```

```
# Note: I use here as a wrapper on the directory to the file
```

```
# because it's safer when the document is knitted.
```

```
personality <- read_csv(here("Data/TIPI_data.csv"))
```

```
## Rows: 1799 Columns: 17
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (9): education, urban, gender, engnat, religion, orientation, race, vote...
```

```
## dbl (8): ID, Extraversion, Neuroticism, Conscientiousness, Agreeableness, Op...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Inspecting data is also a good idea, to ensure the data was read in properly.

```
# Inspect data
str(personality)
```

```
## spc_tbl_ [1,799 x 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ ID : num [1:1799] 1 2 3 4 5 6 7 8 9 10 ...
## $ Extraversion : num [1:1799] 3.5 5.5 4 2.5 2 4 3.5 1 4.5 1 ...
## $ Neuroticism : num [1:1799] 1.5 7 4 2.5 2.5 3.5 1.5 1 2 6.5 ...
## $ Conscientiousness: num [1:1799] 6 4.5 7 5 4.5 4.5 6 3 5 1 ...
## $ Agreeableness : num [1:1799] 6 4 3 5 2.5 4 4.5 4 6.5 7 ...
## $ Openness : num [1:1799] 6.5 6.5 7 5.5 4 7 5 4.5 6 6 ...
## $ education : chr [1:1799] "Univ" "<HS" "Univ" "HS" ...
## $ urban : chr [1:1799] NA "Suburban" "Rural" "Suburban" ...
## $ gender : chr [1:1799] "M" "F" "M" "M" ...
## $ engnat : chr [1:1799] "Non-native" "Native" "Native" "Native" ...
## $ age : num [1:1799] 28 14 25 37 17 23 17 28 32 14 ...
## $ religion : chr [1:1799] "Atheist" "Agnostic" "Other" "Atheist" ...
## $ orientation : chr [1:1799] "Heterosexual" "Bisexual" "Heterosexual" "Bisexual" ...
## $ race : chr [1:1799] "Other" "Indig-White" "Indig-White" "Indig-White" ...
## $ voted : chr [1:1799] "No" "No" "Yes" "No" ...
## $ married : chr [1:1799] "Never married" "Never married" "Never married" "Currently married" ...
## $ familysize : num [1:1799] 1 1 3 2 2 3 2 3 3 2 ...
## - attr(*, "spec")=
## .. cols(
## .. ID = col_double(),
## .. Extraversion = col_double(),
## .. Neuroticism = col_double(),
## .. Conscientiousness = col_double(),
## .. Agreeableness = col_double(),
## .. Openness = col_double(),
## .. education = col_character(),
## .. urban = col_character(),
## .. gender = col_character(),
## .. engnat = col_character(),
## .. age = col_double(),
## .. religion = col_character(),
## .. orientation = col_character(),
## .. race = col_character(),
## .. voted = col_character(),
## .. married = col_character(),
## .. familysize = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
glimpse(personality)
```

```
## Rows: 1,799
## Columns: 17
## $ ID <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ Extraversion <dbl> 3.5, 5.5, 4.0, 2.5, 2.0, 4.0, 3.5, 1.0, 4.5, 1.0, 1.~
## $ Neuroticism <dbl> 1.5, 7.0, 4.0, 2.5, 2.5, 3.5, 1.5, 1.0, 2.0, 6.5, 2.~
## $ Conscientiousness <dbl> 6.0, 4.5, 7.0, 5.0, 4.5, 4.5, 6.0, 3.0, 5.0, 1.0, 3.~
```

```
## $ Agreeableness    <dbl> 6.0, 4.0, 3.0, 5.0, 2.5, 4.0, 4.5, 4.0, 6.5, 7.0, 1.~
## $ Openness         <dbl> 6.5, 6.5, 7.0, 5.5, 4.0, 7.0, 5.0, 4.5, 6.0, 6.0, 4.~
## $ education        <chr> "Univ", "<HS", "Univ", "HS", "HS", "HS", "<HS", "Uni~
## $ urban            <chr> NA, "Suburban", "Rural", "Suburban", "Suburban", "Ru~
## $ gender           <chr> "M", "F", "M", "M", "M", "M", "M", "M", "M", "F", "M~
## $ engnat           <chr> "Non-native", "Native", "Native", "Native", "Native"~
## $ age              <dbl> 28, 14, 25, 37, 17, 23, 17, 28, 32, 14, 16, 35, 18, ~
## $ religion         <chr> "Atheist", "Agnostic", "Other", "Atheist", "Agnostic~
## $ orientation      <chr> "Heterosexual", "Bisexual", "Heterosexual", "Bisexua~
## $ race             <chr> "Other", "Indig-White", "Indig-White", "Indig-White"~
## $ voted            <chr> "No", "No", "Yes", "No", "No", "No", "No", "Yes", "N~
## $ married          <chr> "Never married", "Never married", "Never married", "~
## $ familysize       <dbl> 1, 1, 3, 2, 2, 3, 2, 3, 3, 2, 2, 3, 3, 2, 5, 6, 3, 3~
```

2) Using the personality dataframe, create a new dataframe called `personality_domains` that contains only ID and the five personality domain variables.

```
# Select vars of interest
personality_domains <- personality |>
  select(ID, Extraversion:Openness)

# Check if selection worked
colnames(personality_domains)
```

```
## [1] "ID"                "Extraversion"      "Neuroticism"
## [4] "Conscientiousness" "Agreeableness"     "Openness"
```

3) Using the personality dataframe, create a new dataframe called `hs_or_less` that contains all variables for individuals whose education level is either less than high school or high school. It's wise to first view the unique levels of education

```
# View unique values of education
table(personality$education)
```

```
##
## <HS Grad   HS Univ
## 278   245   757   488
```

Based on this, education takes on four values: <HS, Grad, HS, and Univ. Thus, our filter should ensure that people are either in the “HS” or “<HS” groups.

```
# Filter data so people are in HS or lower
# (Using %in% is Gabe's preference.)
hs_or_less <- personality |>
  filter(education %in% c("<HS", "HS"))
# (The following is equivalent to the above and would also be accepted.)
hs_or_less <- personality |>
  filter(education == "HS" | education == "<HS")

# Check that filter worked
table(hs_or_less$education)
```

```
##
## <HS HS
## 278 757
```

4) Using the personality dataframe, create a new dataframe called married_suburban that contains all variables for individuals who are currently married and live in a suburban neighbourhood.

```
# Check levels of married and neighborhood
table(personality$married)
```

```
##
## Currently married      Never married Previously married
##           314           1383           93
```

```
table(personality$urban)
```

```
##
## Rural Suburban Urban
##    317    815    645
```

```
# Filter data so only currently married people
# in suburban homes appear
```

```
married_suburban <- personality |>
  filter(married == "Currently married" &
         urban == "Suburban")
```

```
# Check that filter worked
# Since only one level appeared,
# the filter worked appropriately
```

```
table(married_suburban$married,
      married_suburban$urban)
```

```
##
## Suburban
## Currently married 141
```

The transformation worked since only one combination of both variables was counted.

5) In the personality dataframe, use familysize to create a new categorical variable called familysize_cat with two categories: 2 or less and more than 2. Hint: You will need to use an ifelse() statement as part of your code.

```
# Check levels of familysize
table(personality$familysize)
```

```
##
## 0  1  2  3  4  5  6  7  8  9 14 15
## 32 274 749 453 169 71 28 16 4 1 1 1
```

```

# Save into new data object because over-writing
# data is poor practice
personality_revised <- personality |>
  mutate(familysize_cat = ifelse(
    familysize <= 2, # Condition
    "Two or less", # Value if TRUE
    "More than two" # Value if FALSE
  ))

# Check if transformation worked
table(personality_revised$familysize_cat)

```

```

##
## More than two    Two or less
##              744          1055

```

```

# OPTIONAL: More thorough check
personality_revised |>
  count(familysize, familysize_cat)

```

```

## # A tibble: 12 x 3
##   familysize familysize_cat     n
##   <dbl> <chr>           <int>
## 1         0 Two or less       32
## 2         1 Two or less      274
## 3         2 Two or less      749
## 4         3 More than two    453
## 5         4 More than two    169
## 6         5 More than two     71
## 7         6 More than two     28
## 8         7 More than two     16
## 9         8 More than two      4
## 10        9 More than two      1
## 11        14 More than two      1
## 12        15 More than two      1

```

A more elegant solution, IMO, is to use `case_when()` instead of `ifelse()`, as it's more flexible in case missing values arise.

```

# OPTIONAL: case_when()
personality_revised <- personality |>
  # Must convert familysize to numeric first
  mutate(familysize = as.numeric(familysize)) |>
  mutate(familysize_cat = case_when(
    # Condition 1 ~ Value if Condition 1 is TRUE,
    # Condition 2 ~ Value if Condition 2 is TRUE,
    # etc...
    familysize <= 2 ~ "Two or less",
    familysize > 2 ~ "More than two",
    # TRUE denotes what to do if all prior logical
    # statements are FALSE (e.g., in case of
    # missing data)
  ))

```

```
TRUE ~ NA_character_  
)
```

6) Using the personality dataframe, compute the mean Agreeableness score separately for individuals who voted and individuals who did not vote.

```
# Check levels of voting  
table(personality$voted)
```

```
##  
##   No   Yes  
## 1205  575
```

```
# Create data frames for people who voted  
# vs. didn't vote  
voted <- personality |>  
  filter(voted == "Yes")  
did_not_vote <- personality |>  
  filter(voted == "No")  
  
# Check if filters worked  
table(voted$voted)
```

```
##  
## Yes  
## 575
```

```
table(did_not_vote$voted)
```

```
##  
##   No  
## 1205
```

```
# Compute means for both data frames  
mean(voted$Agreeableness)
```

```
## [1] 4.445217
```

```
mean(did_not_vote$Agreeableness)
```

```
## [1] 4.231535
```

The mean Agreeableness was higher among those who voted ($M = 4.45$) than those who did not vote ($M = 4.23$), though this mean difference was small.